

Rational Functions & Limits

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **identify** holes and vertical, horizontal, or slant asymptotes of a rational function.

RELATED SKILLS

☐ I Determine horizontal or slant asymptotic behavior.☐ I Determine vertical asymptotes from noncanceling denominator zeros.☐ I Identify a removable discontinuity and its coordinates.☐ I Recall and correctly use rational-function terms including restriction, hole, asymptote, and end behavior.☐ A Cancel common factors rather than individual terms.☐ A Factor numerators and denominators completely.☐ R Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.

☐ 2. I can **sketch and analyze** a rational function using its algebraic and asymptotic features.

RELATED SKILLS

☐ I Determine the sign of a rational function on intervals.☐ D Determine horizontal or slant asymptotic behavior.☐ D Determine vertical asymptotes from noncanceling denominator zeros.☐ D Identify a removable discontinuity and its coordinates.☐ A Determine excluded values from an original denominator.

☐ 3. I can **interpret** limit notation as a statement about function behavior near an input.

RELATED SKILLS

☐ I Read and write limit notation with correct approach direction.

☐ 4. I can **estimate** a limit from a graph or table.

RELATED SKILLS

☐ I Estimate a limit from input-output values approaching from both sides.☐ I Estimate one-sided and two-sided limits from a graph.

☐ 5. I can **evaluate** a limit algebraically using direct substitution or simplification.

RELATED SKILLS

☐ I Resolve an indeterminate algebraic form by factoring or rationalizing.☐ I Test a limit using direct substitution.☐ A Factor numerators and denominators completely.

☐ **6. I can **interpret** an infinite limit as vertical-asymptotic behavior.**

RELATED SKILLS

- ☐ I Interpret unbounded output near a finite input as an infinite limit.
 - ☐ D Determine vertical asymptotes from noncanceling denominator zeros.
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☐ **7. I can **use** limit notation to describe behavior at infinity.**

RELATED SKILLS

- ☐ I Compare dominant terms to determine behavior at infinity.
 - ☐ D Determine horizontal or slant asymptotic behavior.
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☐ **8. I can **analyze** the long-run behavior of quotients involving polynomial, exponential, or logarithmic expressions.**

RELATED SKILLS

- ☐ D Compare the long-run growth or decay of functions from different families.
 - ☐ A Compare dominant terms to determine behavior at infinity.
 - ☐ A Compare functions shown in different representations.
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☐ **9. I can **compare** the long-run growth of polynomial, exponential, logarithmic, and rational functions.**

RELATED SKILLS

- ☐ I Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions.
 - ☐ A Compare functions shown in different representations.
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☐ **10. I can **build or select** a rational or rational-adjacent model for a context and interpret its long-run behavior.**

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
 - ☐ D Select a function family whose behavior fits a table, graph, or context.
 - ☐ A Determine horizontal or slant asymptotic behavior.
 - ☐ A Restrict a model to meaningful contextual inputs.
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